

The Path to Automated Narrative Processing: Narrativity, Uncertainty, and Slow Feature Analysis

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Abstract

How does narrative figure into the drive towards artificial intelligences that bear comparison to human cognition? There is a robust literature about narrative computation, focusing on the analysis of narrative structures, as well as large language models' ability to generate narratives. While these directions automate two high-level human cognitive activities, the processes do not resemble how narrative structures are generated and processed into abstractions and memories in human thinking. This presentation describes the function of ongoing narrative processing in low-level cognition, followed by consideration of two robust areas of research that might advance our understanding and anticipate the role it might play in artificial intelligence. The presentation begins by considering the problem ongoing narrative processing helps us solve. I will review and update a model of narrativity as an emergent feature in event processing predicated on parsing event boundaries via extant cognitive models, uncertainty reduction, affect, and attributing meaning (Miller, 2024). I then focus on two key, but limited, topics within this field. In cognitive science, new models of decision-making theorize a key role for uncertainty in time segmentation and abstraction in event processing (Wang et al., 2025). In artificial intelligence, 'slow feature analysis' works to identify event constituents and define event boundaries (Song and Zhao, 2022). I will argue for relationships between these processes and emergent narrativity in event processing. Taken together, these perspectives contribute to research directions in cognitive studies, artificial intelligence, and narrative theory around the question of what human-like event processing would look like in AI. Finally, I consider embodiment, processing limitations, and the ethics of this research direction.

Keywords: narrative processing, artificial intelligence, event cognition, AGI, narrative computation

1 Introduction

How does narrative figure into the drive towards artificial intelligences that bear comparison to human cognition?

There is a robust literature about narrative computation, focusing on the analysis of narrative structures. Large language models can be leveraged to generate narratives. While these two

directions automate two high-level human cognitive activities, the processes do not resemble how narrative structures are recognized and created in human thinking.

This article describes the function of ongoing narrative processing in human intelligence, followed by consideration of the state of artificial intelligence in approximating that functionality.

2 The Problem Narrative Processing Solves

Ongoing narrative processing helps us solve a fundamental cognitive problem: orienting ourselves towards, framing responses to, and retroactively parsing experience.

2.1 Orienting Towards Experience

2.2 Framing Responses

2.3 Retroactive Parsing

3 Human Event Processing

4 AI Event Processing: State of the Art

4.1 Narrative Analysis Systems

4.2 Narrative Generation

4.3 Gaps and Limitations

5 A Research Roadmap

5.1 Direction 1

5.2 Direction 2

5.3 Contributions to AGI

6 Ethics and Open Questions

6.1 Is Embodiment Essential?

Is embodiment an essential precondition of narrative processing?

6.2 Limitation or Feature?

Is the “functionality” of narrative processing actually a limitation imposed by processing capacity, and might artificial systems ultimately transcend that limitation?

7 Conclusion

References

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